Link to Google Doc

Milestone 1

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Milestone 2

Who is your audience? Come up with at least three options and pick one target audience.

For this project, we examine how cities market themselves through tourism websites by analyzing the adjectives, phrases, themes, slogans, and other language official tourism organizations use to construct a city's identity. By comparing this language across cities, we hope to uncover which cities present themselves similarly, which cities have distinctive identities, and which themes dominate tourism marketing more broadly.

With our topic in hand, we came up with the following prospective audiences:

- **Curious Travelers:** People deciding where they would like to travel who are interested in understanding the perceived personality or identity of different destinations.
- **Tourism and destination-marketing professionals:** People who work for tourism boards, city governments, advertising agencies, or destination-marketing organizations and want to understand how their city's branding compares with competitors.
- **City enthusiasts, students, and researchers -** People interested in cities, geography, culture, linguistics, urban studies, or marketing who want to understand how urban identities are constructed through language.

Final Selection:

We ultimately decided our target audience is prospective travelers exploring and comparing potential destinations. These travelers may already have general impressions of different cities, but they want to better understand what distinguishes one destination from another. By showing how cities describe and market themselves, our visualization can help travelers identify destinations that emphasize experiences they care about, such

as food, nightlife, history, culture, or outdoor activities, and discover cities that may be more similar than they initially expected.

Describe your target audience in more detail. What do they know? What are their interests? What visualization literacy do they have? At what level of detail will you present information to them?

Noted above as curious travelers and city enthusiasts, our target audience will likely be general-interest readers rather than experts. They can recognize cities through personal experience and recognize destination stereotypes (Paris is romantic, Las Vegas is exciting, etc.). Their interests may center on comparison and discovery: which cities market themselves alike, whether their own city stands out, and what the tourism boards promise.

Our audience's visualization literacy is probably moderate. They are comfortable with charts common in journalism: maps, bar charts, word clouds, simple scatter plots, but not the more specialized forms, so any unfamiliar visualization will need clear labeling and an introduction.

We will therefore present information with headlines and clear takeaways, letting users drill down into individual cities and comparisons on demand, while keeping methodological detail in a brief footnote.

What questions about your data will be interesting for your audience? Come up with a list of interesting questions that your audience may have about your data. The more, the better, but your team should come up with at least ten questions.

Potential audience questions

- What are major tourist cities trying to be known for? What words and phrases do cities use most often to market themselves?
- Which cities use the most similar language to describe themselves?
- Which cities have the most distinctive or unique marketing language?
- Do major tourist cities rely on the same adjectives and phrases to attract visitors?
- How do "hidden gem" cities market themselves differently from major tourist destinations?
- What themes do cities emphasize most?
- Are cities competing to make the same superlative claims about themselves?
- How do city marketing languages compare to actual reviews?

Are there cities that seem very different geographically but market themselves almost identically?

Can we identify a city based only on the words and phrases used on its tourism website?

What data do you have or plan to obtain? Briefly describe the data you envision to use and the respective data types (categorical, ordinal, or quantitative) in your process book. It's OK if you are unsure about the data type for some attributes - you can simply describe them (e.g., geographic location).

Data

For the data, we plan to compile a dataset of ~25 major tourist cities, including each city's name, geographic location, and country (categorical). In addition, we may compare 25 major tourist cities with 25 lesser-known "hidden gem" cities to examine whether their tourism marketing strategies differ. For each city, we will scrape its official tourism website, collecting the language used to market the destination, including adjectives, recurring phrases, slogans, mottos, and descriptions of attractions (categorical). We will also record quantitative measures such as word frequency, word count, and frequency of recurring themes.

To compare cities, we plan to categorize marketing language into broader themes such as history, culture, food, nature, nightlife, luxury, adventure, etc. (categorical). We may also calculate measures such as language similarity between cities, allowing us to identify which cities market themselves in surprisingly similar or distinct ways. Geographic location will allow us to examine whether cities in the same region tend to market themselves similarly. We can also compare cities' use of superlatives to understand how they use claims of uniqueness to attract tourists.

Process

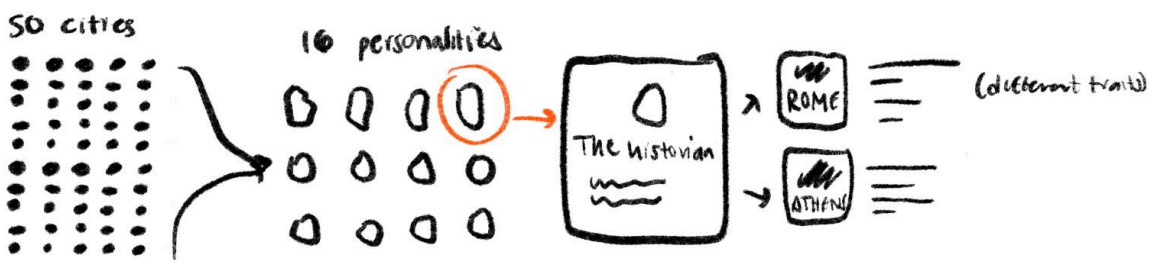
To develop our final project topic, we each independently brainstormed 3–5 ideas in a shared Google Sheet, keeping the initial idea space broad. As a group, we agreed that we wanted to pursue a lighthearted and accessible story that could appeal to a general audience while still offering meaningful insights. We then came together to rank each idea on a 1–5 scale (5 being the highest) based on originality, storytelling potential, and data availability. After comparing our scores and discussing our top 3 candidates, we ultimately selected our final project idea.

A	B	C	D	E	F
Nate	**Rating (1-5)**	**Josh**	**Rating (1-5)**	**Christine**	**Rating 1-5**
How **Flight Routes** changed throughout the years	3	Which **restaurant items** provide the best macronutrient bre	4.5	How tourist websites describe their cities; how similar are t	5
Dog Breeds throughout the years	3.5	Every Species/scientific/street names/thing named after hu	4	What People Ask the Internet at a specific time of day	3.5
Opening Meta throughout history using lichess data	4.5	What is most associated with each number/What numbers	3.5	How different countries around the world percieve colors	3
				What $XX could get you in different places	3

Sketches

Christine

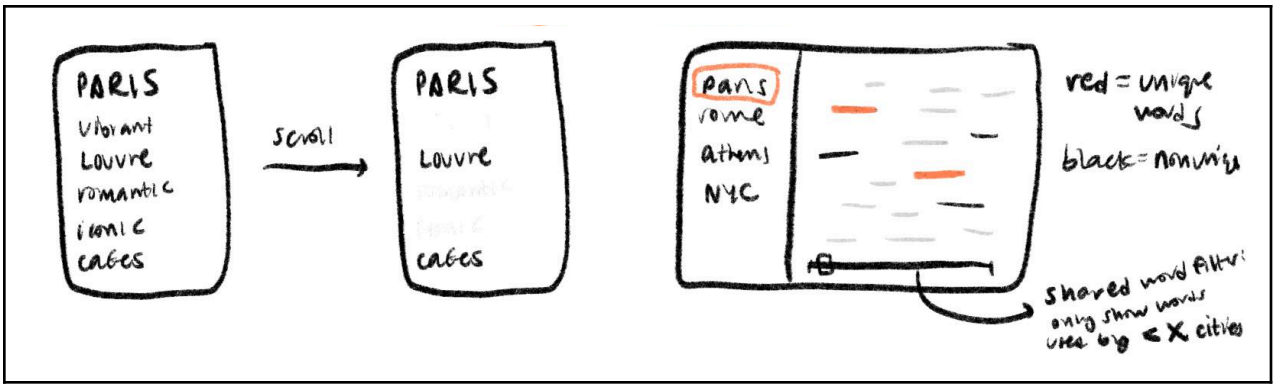
(Reviewed as Sketch #1) Sketch #1: Which cities use the most similar language to describe themselves?
Based on their most common adjectives and phrases, each city is given a "personality," similar to an MBTI profile. Users can explore cities to compare their personalities and see which destinations are most alike.



(Reviewed as Sketch #2) Sketch #2: Are cities competing to make the same superlative claims about themselves?
A city matchup where users compare competing claims such as "most stunning natural wonders" or "best nightlife" and decide which city makes the stronger case. This is geared toward more well-traveled audiences.



(Reviewed as Sketch #3) Sketch #3: Which cities have the most distinctive or unique marketing language?
Each city is given an "identity card" with excerpts taken from its website. As the user scrolls or interacts, generic tourism clichés disappear, revealing the language that makes it unique. There can also be a larger visualization where users can see which words are associated with certain cities and filter words to only show the ones used by < X cities.



Reflection

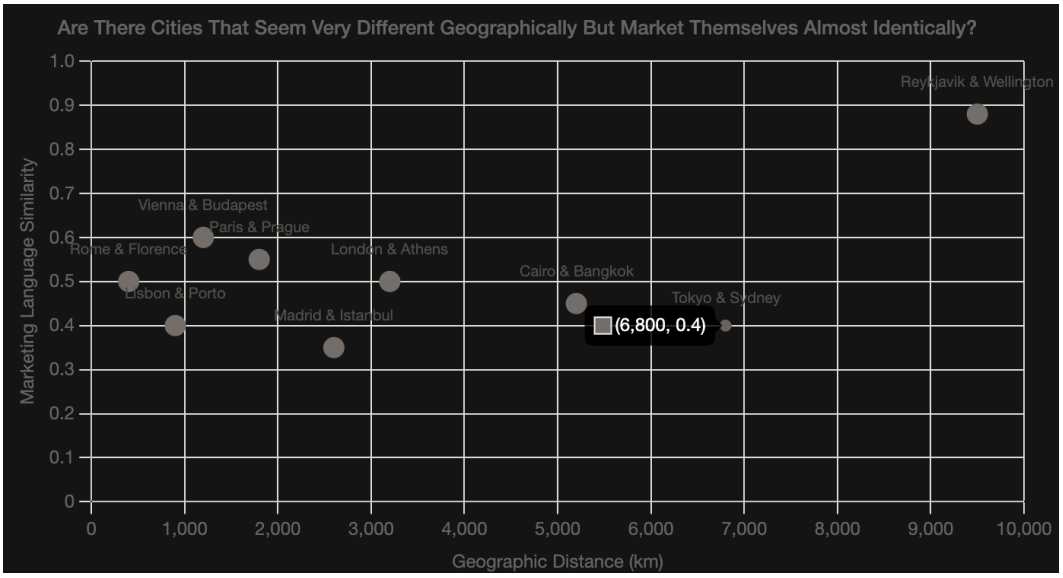
For my sketches, I decided to stick with the original questions we settled on because they were formed around the dataset we are planning to obtain. However, as I worked through the sketches, I realized that some questions allow for much richer and more multi-dimensional exploration than others, and that several of our questions can naturally be combined under a larger question. For instance, “Which cities use the most similar language to describe themselves?” naturally follows from “What words and phrases do cities use most often to market themselves?” The first question adds another layer to the second, since simply showing the most common words could easily become a basic word-bubble visualization, while comparing cities based on those words creates more opportunities for interaction and storytelling. I also noticed that many of my sketches ended up addressing multiple questions at once, which made me think more about how the questions can work together rather than being treated as separate visualizations. I anticipate that once we obtain our final dataset, our questions will evolve even further, as we will have a more nuanced understanding of the patterns in the data and can let those patterns guide the story.

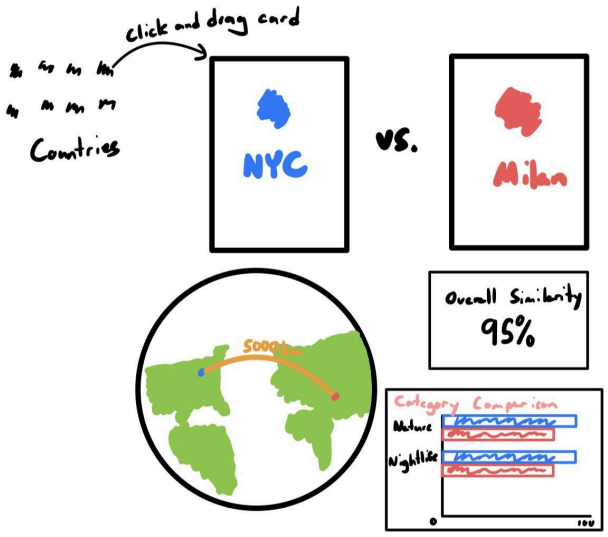
Josh

(Reviewed as Sketch #4 & #5) Sketch #1: Are there cities that market themselves very differently geographically, but market themselves identically?

Image 1: A scatter plot maps geographic distance against marketing language similarity for city pairs, making the overall trend visible at a glance.

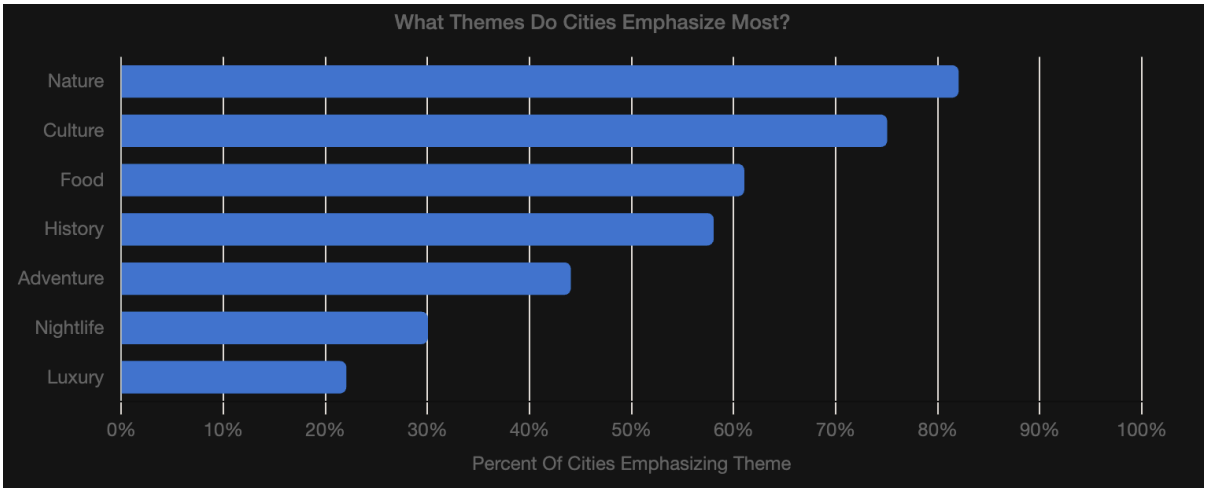
Image 2: Clicking any data point on the overall trend scatter plot opens a deeper comparison: a draggable card map showing the two cities and the distance between them, an overall similarity score, and a category-by-category breakdown of shared traits like nature and nightlife. Alternatively, we can make the deep-dive scenario a standalone visualization.





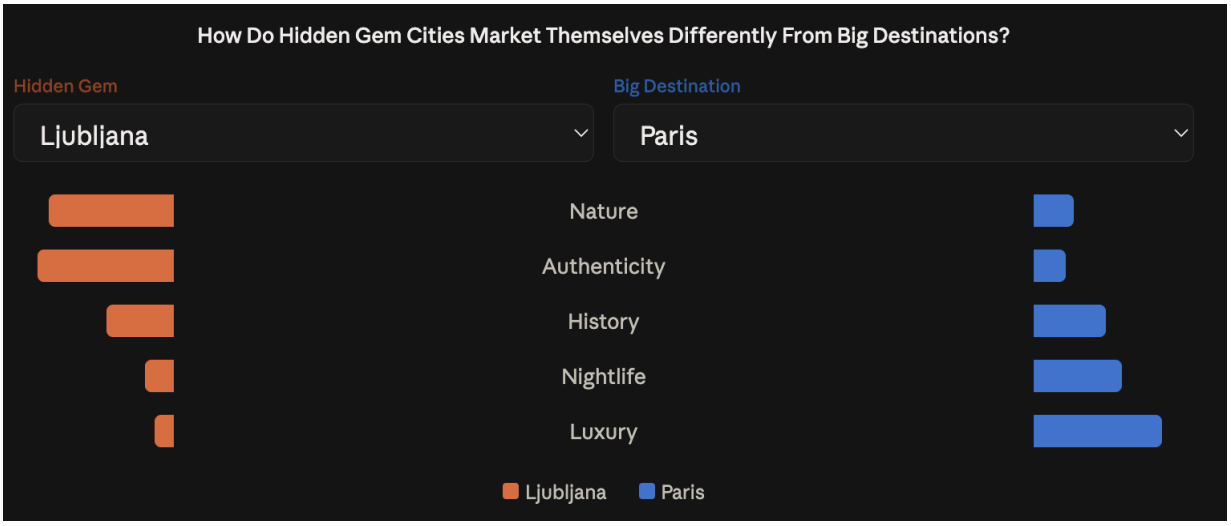
(Reviewed as Sketch #6) Sketch #2: What Themes Do Cities Emphasize the Most?

A horizontal bar chart ranks marketing themes by how often they appear across all city tourism sites, showing at a glance which themes dominate the industry as a whole rather than any single city's strategy.



(Reviewed as Sketch #7) Sketch #3: How do "hidden gem" cities market themselves differently from major tourist destinations?

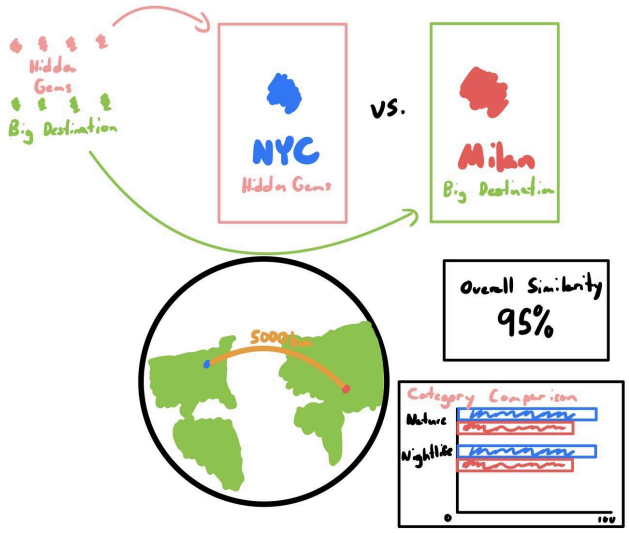
A diverging bar chart lets users pick any hidden-gem city and any major destination from two dropdowns, splitting marketing themes down the middle to compare that specific pair.



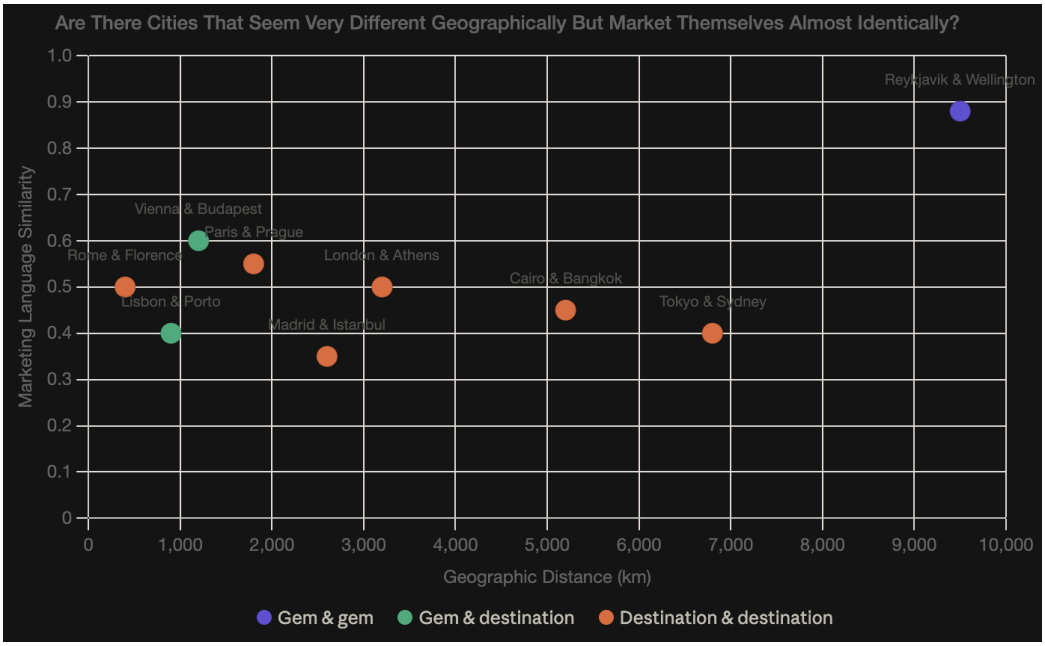
Reflection

For my sketches, I generally stuck with the original questions that my team and I developed. This is mainly because we had already met to plan how to scrape data and what kind of data we needed for this project.

However, working through some of these problems made me realize how much overlap exists between the solutions to the questions we had been treating independently and how they all fit together. The clearest example is in sketches 1 and 3. After sketching the two, I realized that we could simply implement the two together as follows:



(Reviewed as Sketch #8)



(Reviewed as Sketch #9)

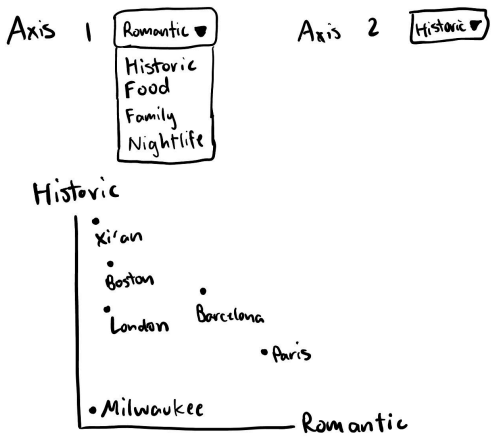
Similar to the detailed comparison view in sketch #1, adding a categorical indicator for hidden gem versus big destination lets the same visualization answer both sketch #1 and sketch #3 with almost no extra UI. The globe, similarity score, and category breakdown stay identical. Meanwhile, the only change is the categorical indicator for what types of cities are being compared. Likewise, in the overview scatter plot itself, coloring each point by whether it's a gem-gem, gem-destination, or destination-destination pair reveals broader trends at a glance, rather than requiring a click into each pair to see the category. This made me realize we should be designing our visualizations around reusable interactions rather than building a separate one for each question, so the final product stays cohesive and doesn't ask the user to relearn a new interface every time they move to a new question. The interface should be as easy and concise to use as possible.

Secondarily, while we have only done isolated sketches so far, we will likely need to figure out the chronological order in which each visualization appears later to maximize user experience as well. I expect the questions we have developed to evolve and potentially expand as we begin pulling our actual dataset.

Nate

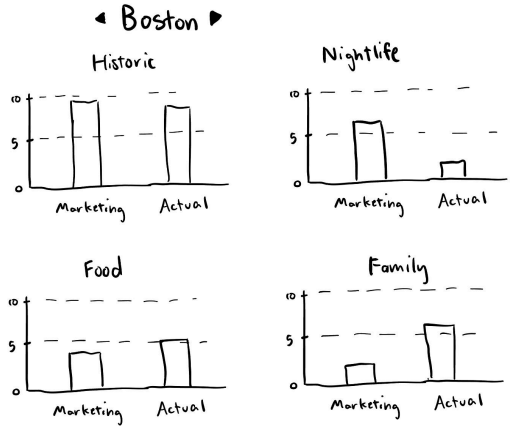
(Reviewed as Sketch #10) Sketch #1: What are the cities known for?
Users select two qualities at a time from a dropdown menu. Some sample selections are included in the drawing. Based on our web-scraped adjectives, each city will have a

number associated with their relevance to each quality. Cities will then be shown on the graph.



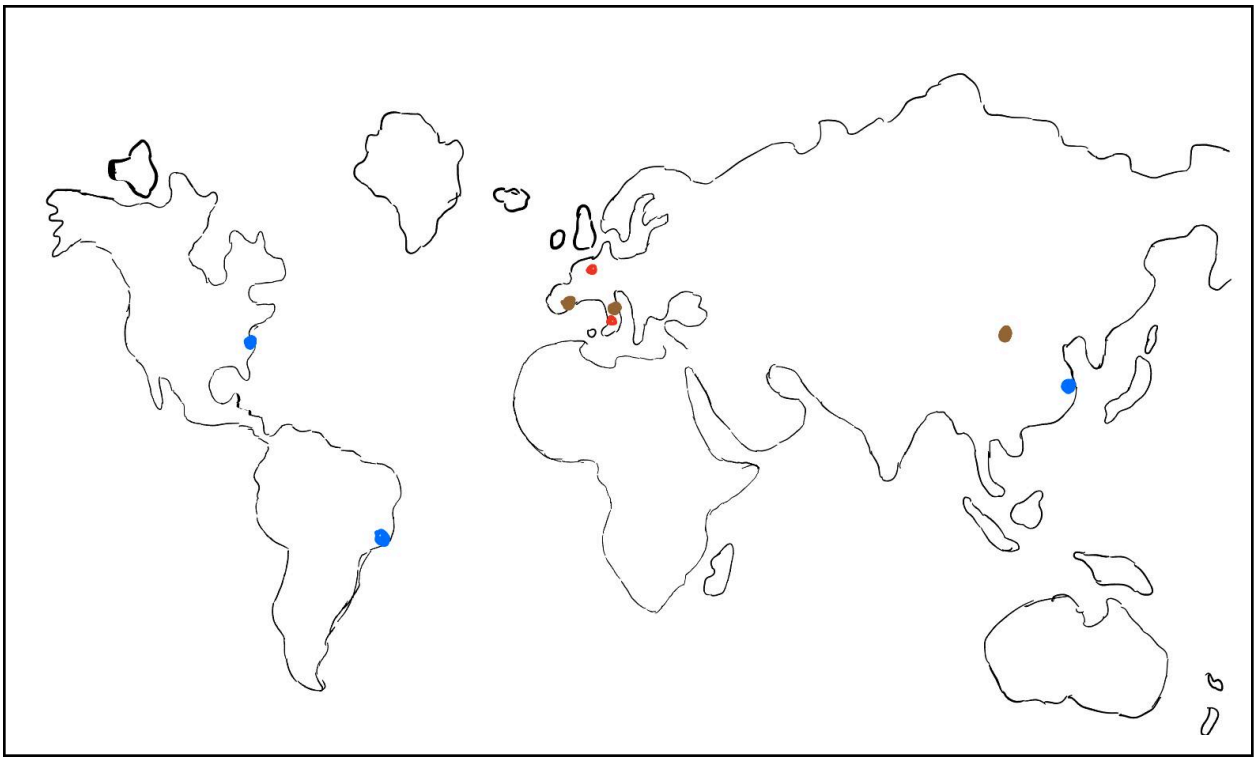
(Reviewed as Sketch #11) Sketch #2: How do city marketing languages compare to actual reviews?

Web-scraped adjectives can potentially be separated into marketing and actual review. Users can scroll through cities and see discrepancies between marketing language and actual reviews.



(Reviewed as Sketch #12) Sketch #3: Are there cities that seem very different geographically but market themselves almost identically?

Cities of interest can be overlaid on top of a world map. Certain adjectives will get different colors, and each city will be colored by their primary adjective.



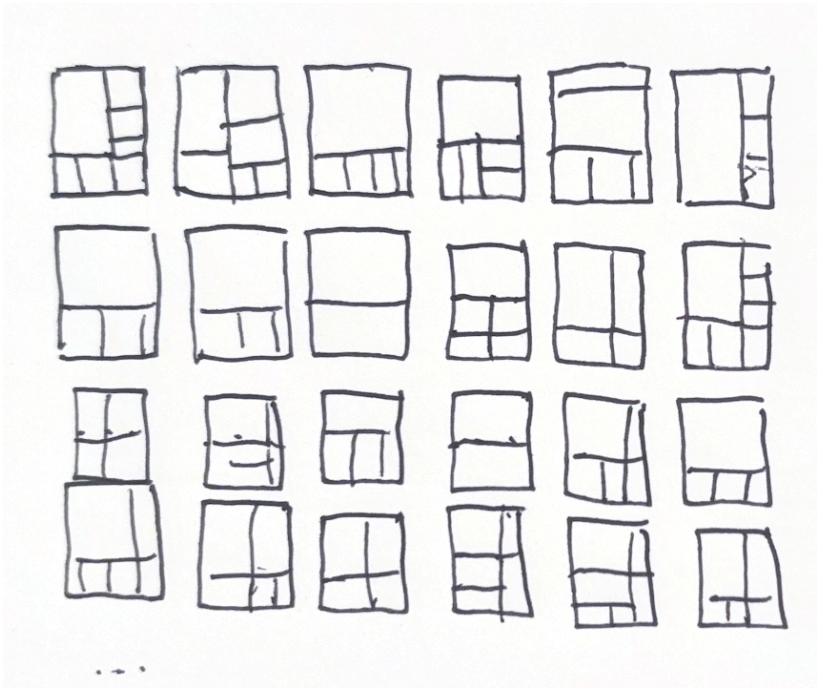
Reflection:

For my sketches, I stayed close to our original team questions, but working through them made me realize that some questions lend themselves to interaction much more naturally than others. My first sketch, which lets users choose two qualities and see where cities fall along those axes, began as "What are cities known for?" but evolved into something more exploratory than the original question implied. Instead of us telling the audience what each city is known for, the user constructs their own comparison, which fits our target audience of curious travelers better. My second sketch addresses our question about how marketing language compares to actual reviews, and sketching it revealed a practical concern: it depends on obtaining a second dataset of review language, so whether we keep it will depend on data availability rather than on the strength of the question itself. My third sketch, mapping cities colored by their primary adjective, made me notice that our geographic questions may be the most visually rewarding ones, since they combine a familiar chart form (a map) with a surprising insight (distant cities sharing identities). Overall, I think the questions that let users make choices: selecting qualities, comparing cities, are stronger for our audience than questions with a single fixed answer, and I expect we will consolidate several of our original questions into these interactive forms once we see the real patterns in our scraped data.

Milestone 3

Christine

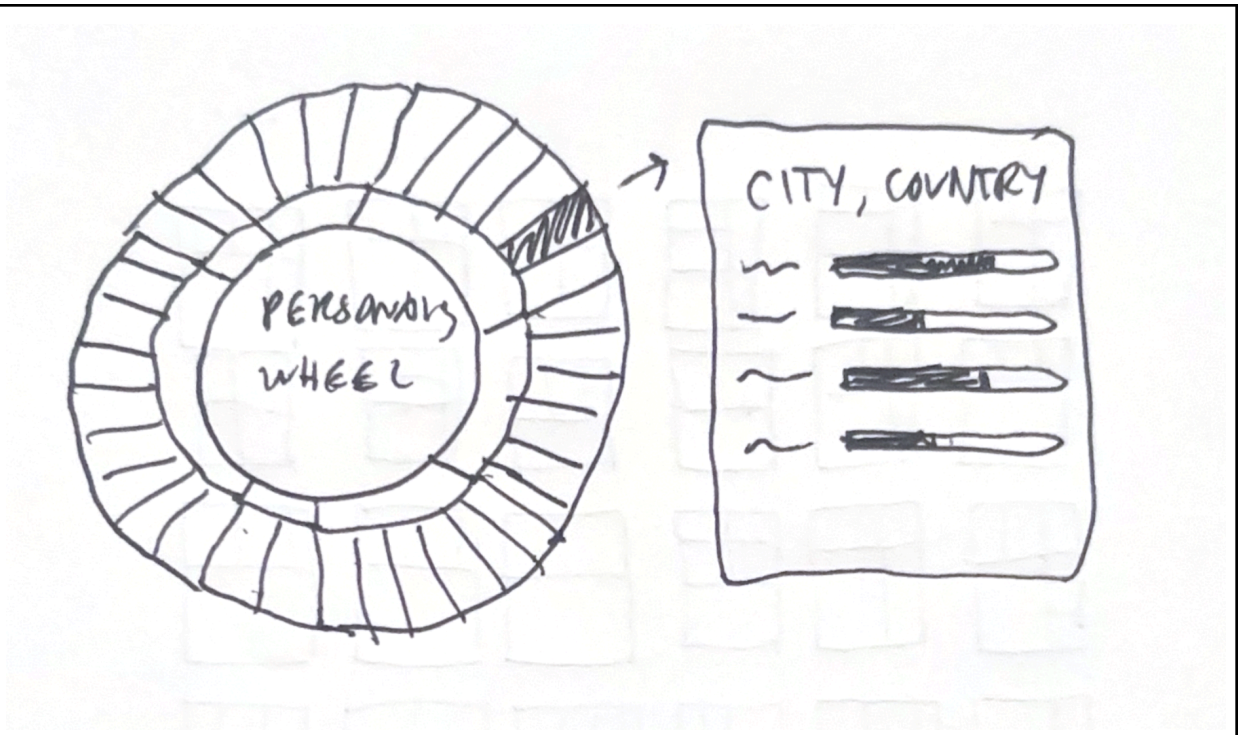
(Reviewed as Sketch #13) Sketch #1
Question: What are cities trying to be known for? What themes do cities emphasize most?



Description: Treemaps give each city a unique "fingerprint" based on the themes they focus on. There will be 10 options: food, history, art & culture, nature, beach, nightlife, shopping, architecture, modern & business, and family.
Legend:

- **Theme - categorical:** color of area plot
- **Theme prevalence - quantitative:** size of area plot

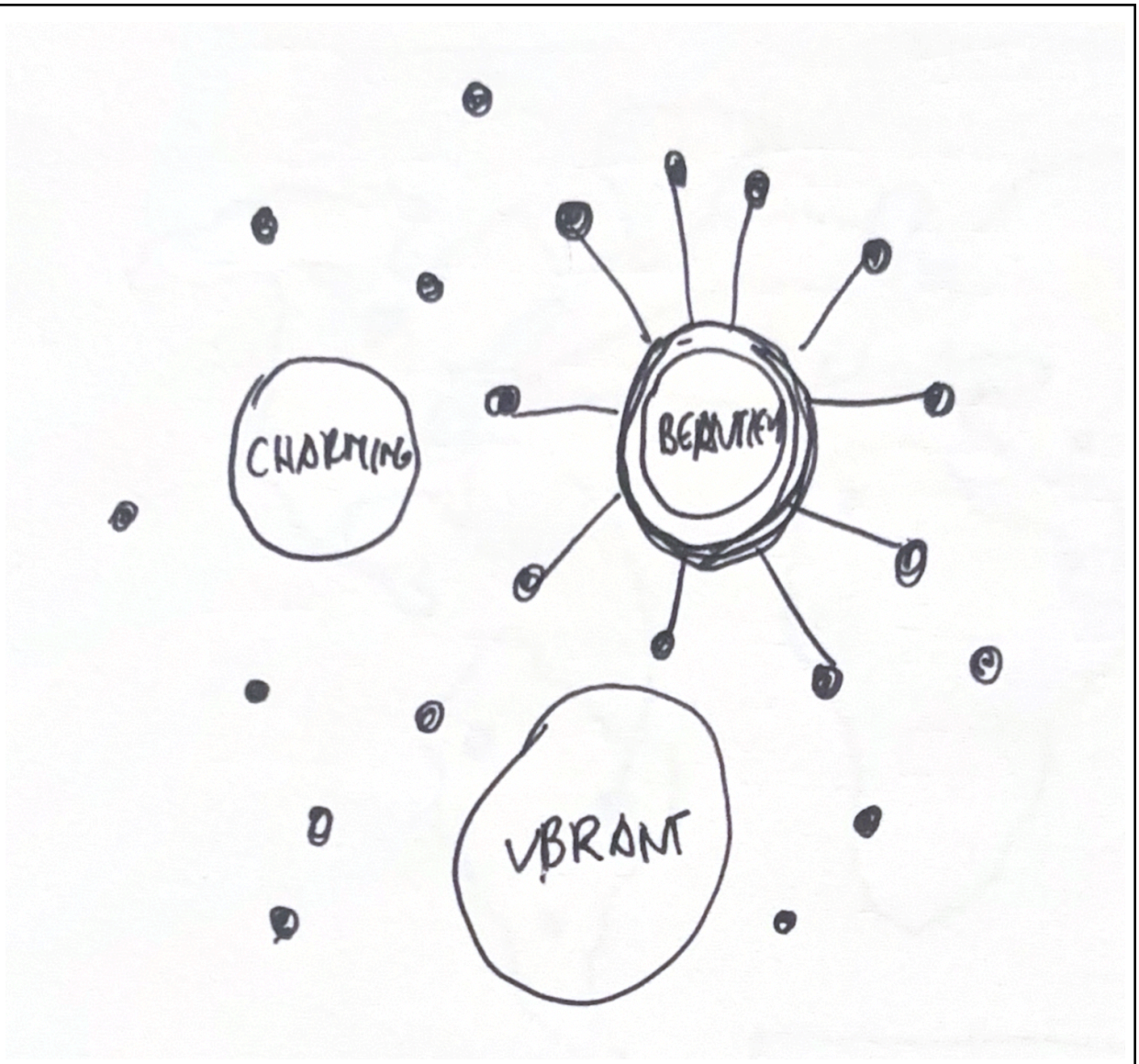
(Reviewed as Sketch #14) Sketch #2
Question: Which cities use the most similar language to describe themselves?



Description: Based on their most common adjectives and phrases, each city is given a "personality," similar to an MBTI profile. The archetypes sit on a sunburst chart, with each wedge sized by how many cities fall into it. Clicking a wedge lists the cities and their defining fingerprint words.
Legend:

- **Archetype - categorical:** wedge color
- **Number of cities per archetype - quantitative:** wedge size
- **Category (popular/hidden) - categorical:** inner vs. outer ring
- **Fingerprint words - categorical:** revealed on click

(Reviewed as Sketch #15) Sketch #3
Question: What words and phrases do cities use most often to market themselves? Do tourist cities rely on the same adjectives and phrases to attract visitors?



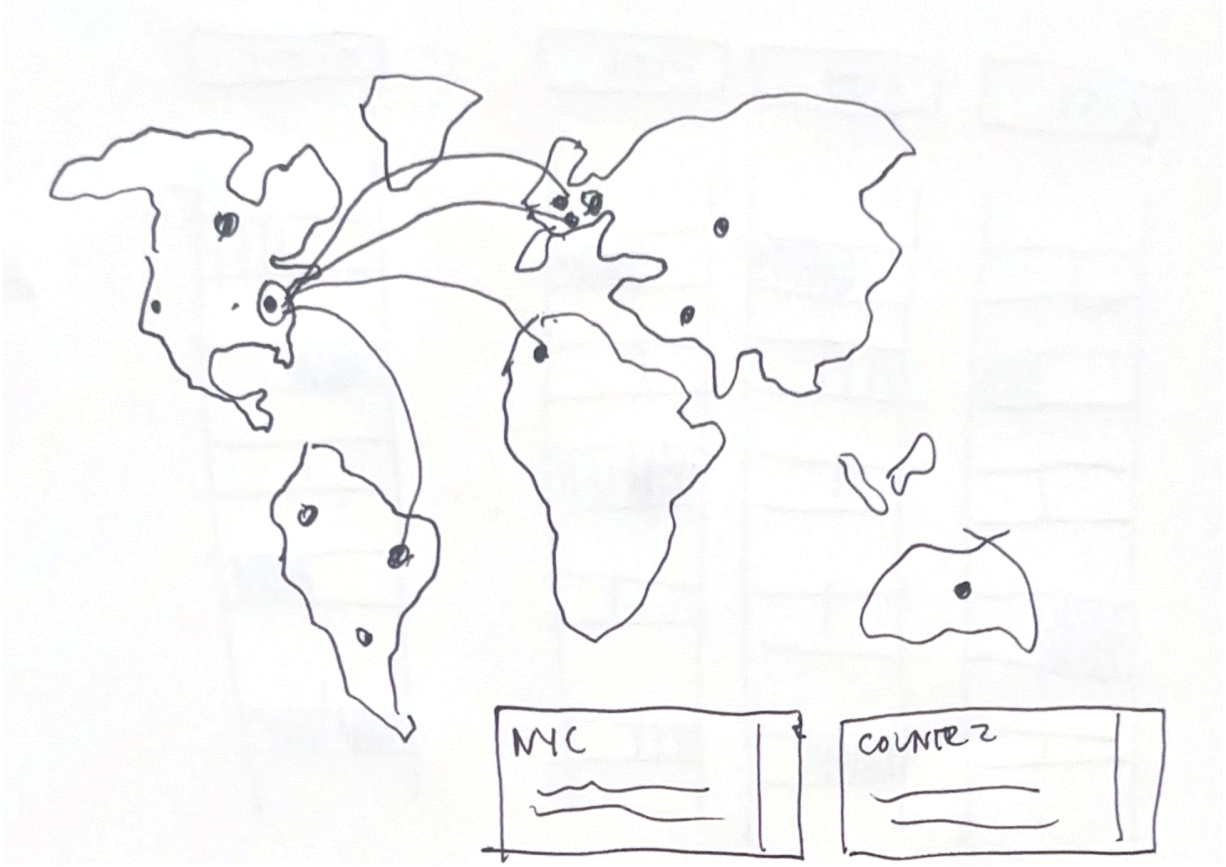
Description: A hub-and-spoke graph where large nodes are shared words and small nodes are cities. Clicking a word draws lines to every city that uses it. A toggle switches the word pool between adjectives, high-sentiment nouns, superlatives, phrases, etc.

Legend:

- **Word type (adjective/noun/superlative/phrase) - categorical:** toggle, one pool shown at a time
- **Word frequency across cities - quantitative:** hub node size
- **City - categorical:** spoke node
- **Word-city usage:** connecting line, drawn on click

(Reviewed as Sketch #16) Sketch #4

Question: Are there cities that seem very different geographically but market themselves almost identically?



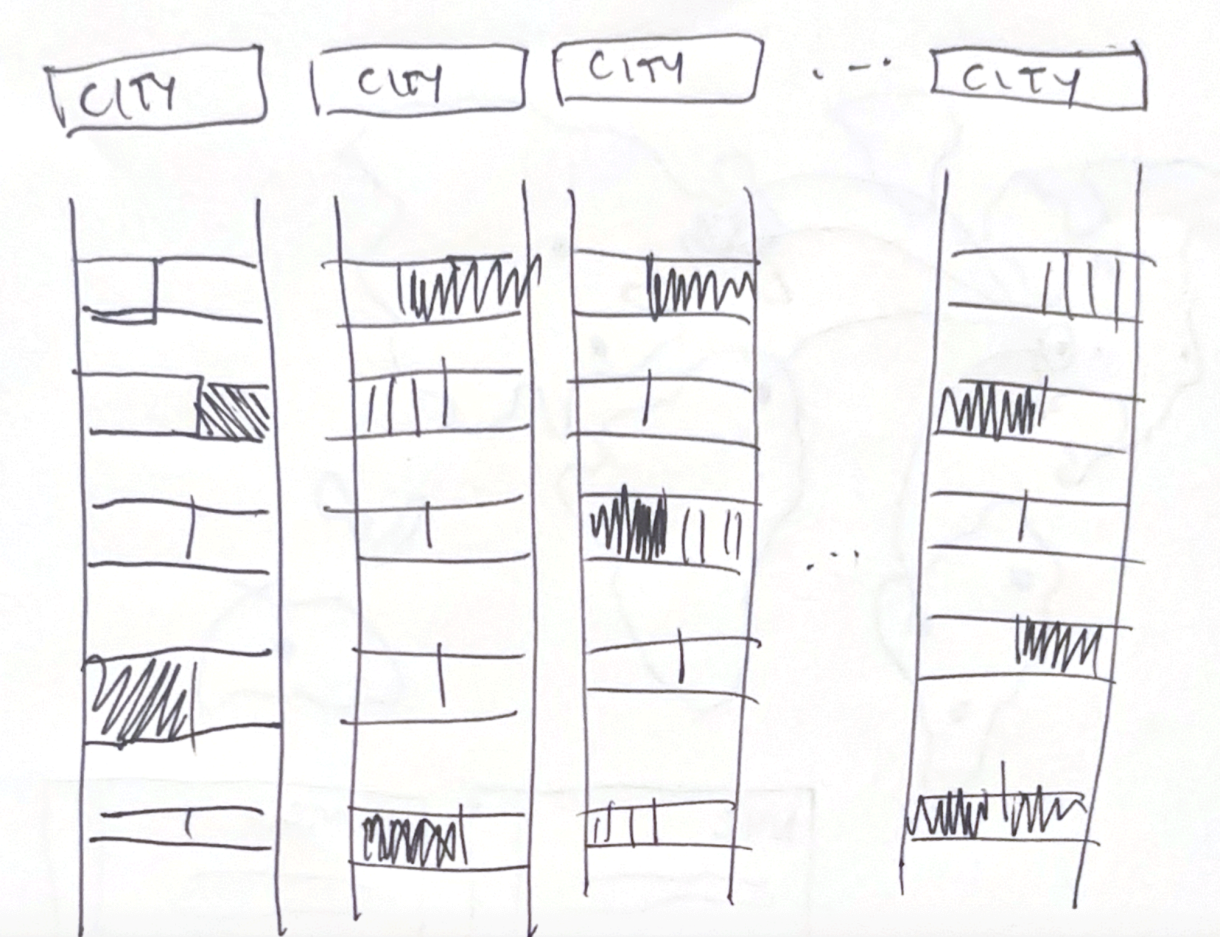
Description: A world map where clicking a city draws arcs to the cities most textually similar to it. Two boarding-pass-style cards appear below allowing users to compare the shared language between the selected pair.

Legend:

- **City location - categorical:** map point
- **Textual similarity - quantitative:** arc thickness
- **Same-region vs. cross-region pair - categorical:** arc color
- **Shared fingerprint words - categorical:** listed on the comparison cards

(Reviewed as Sketch #17) Sketch #5

Question: What are cities trying to be known for? What themes do cities emphasize most?



Description: For each city, its top 10 high-sentiment word pairs render like a DNA strand. There will be linked tiles running down the page, each tile colored by part of speech.

Legend:

- **Part of speech (adjective/noun/superlative) - categorical:** tile color
- **Word-pair rank (most to least sentiment-loaded) - ordinal:** position along the strand
- **City - categorical:** one strand per city

Other sketch ideas

Question: Do the most-visited cities “try” as hard as hidden-gem cities?

Description: A scatter plot comparing how famous a city already is against how much linguistic effort its tourism board puts into sounding distinctive.

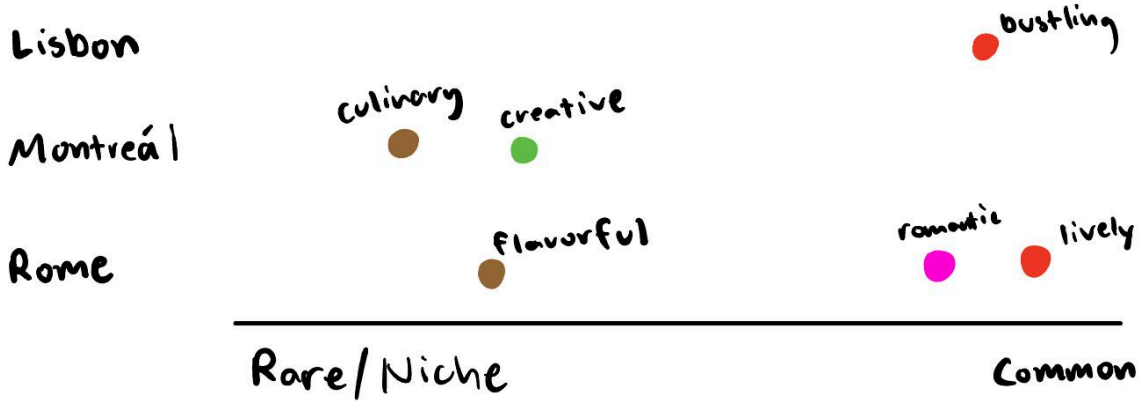
Legend:

- **Annual international tourists - quantitative:** x-axis
- **Distinctiveness or cliché rate - quantitative:** y-axis (toggle between two metrics)
- **Category (popular/hidden) - categorical:** point color

Nate

(Reviewed as Sketch #18) **Sketch #1**

Question: Which cities have the most distinctive or unique marketing language?



Description: Cities are stacked vertically, with their most common descriptors placed along an x-axis. The uniqueness of the descriptor determines its position. The "theme" of the descriptor determines the color. For example,

Legend:

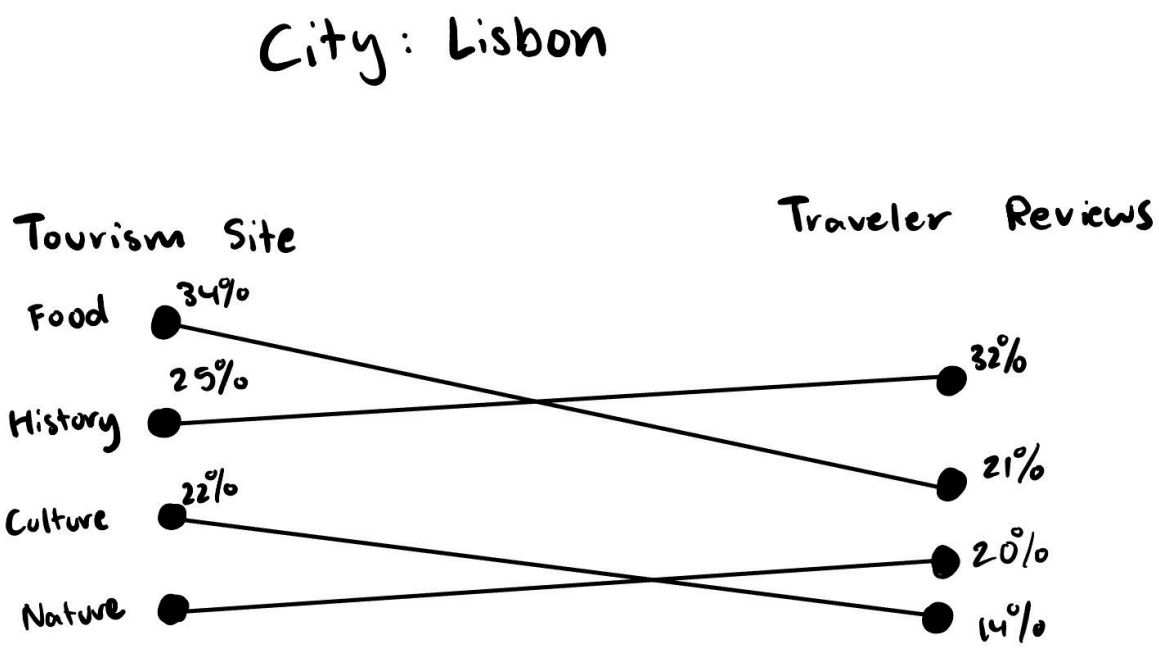
Position: uniqueness of descriptor

Color: "theme" of descriptor

Size (not shown but possible): frequency of descriptor

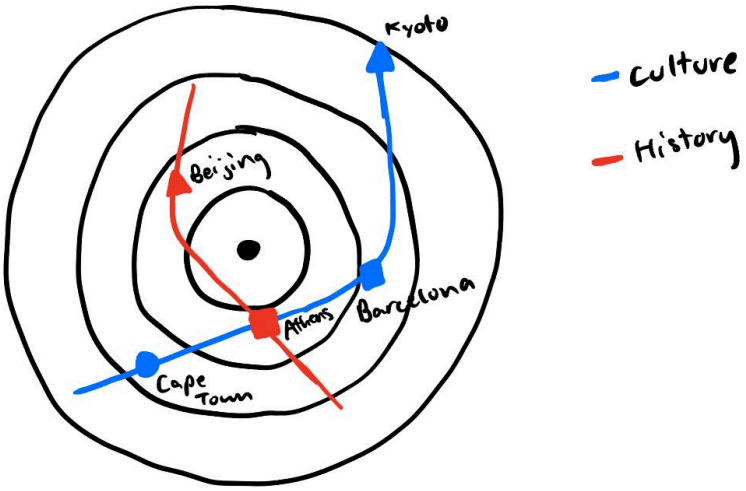
(Reviewed as Sketch #19) **Sketch #2**

Question: How do city marketing languages compare to actual reviews?



Description: For a selected city, users can see the discrepancy between tourism site language and actual review language.
Legend:
Frequency (a metric we will decide on later) is shown for tourism sites on the left and actual traveler reviews on the right. The frequency of each word is shown by the vertical position.

(Reviewed as Sketch #20) Sketch #3
Question: Which cities use the most similar language to describe themselves?



Description: This chart shows how different cities, despite potential large geographical distances, can use similar themes to describe themselves. It is inspired by the distance-based metro map seen in class.

Legend:

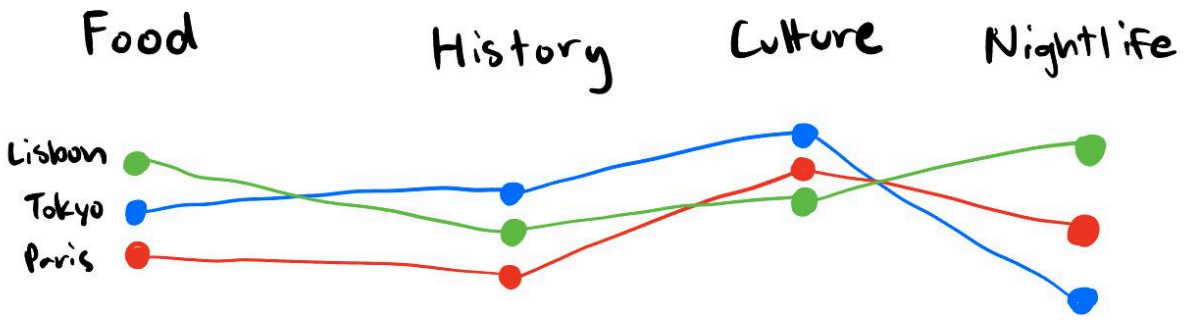
Distance from the center represents how frequent the most common theme is actually used. For example, despite being most frequently described as cities with “culture”, the descriptor still appears more commonly in Barcelona reviews than in Kyoto reviews.

Theme of the most common descriptor is given by color. Cities lying on multiple lines (e.g. Athens) has a similar amount of use for both descriptors.

Shape of the city represents the continent the city lies in.

(Reviewed as Sketch #21) Sketch #4

Question: What are major tourist cities trying to be known for? What words and phrases do cities use most often to market themselves?



Description: Select cities along with descriptors are listed. The relative frequency of descriptor usage for each word is marked by vertical position. It allows users to easily compare what cities are known for relative to others.

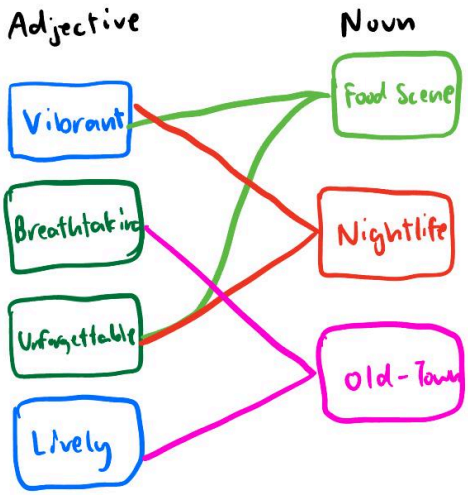
Legend:

Color: each city is assigned a color. Although this is a redundancy, it allows users to more easily track cities across the graph.

Position: represents the frequency of each descriptor

(Reviewed as Sketch #22) Sketch #5

Question: Do major tourist cities rely on the same adjectives and phrases to attract visitors?



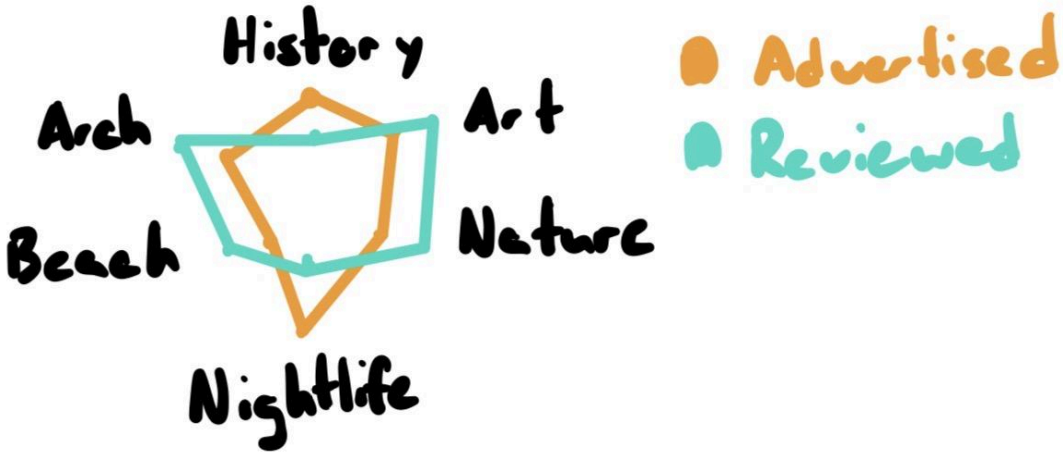
Description: The actual cities are removed from this graphic. Instead, we

focus on the common connections between adjectives and nouns
Legend:
Color: “Theme” of word
Line: common connection between adjective and noun
Boldness of line (not shown but possible): frequency of connection

Josh

(Reviewed as Sketch #23) Sketch #1

Question: How do city marketing languages compare to actual reviews?



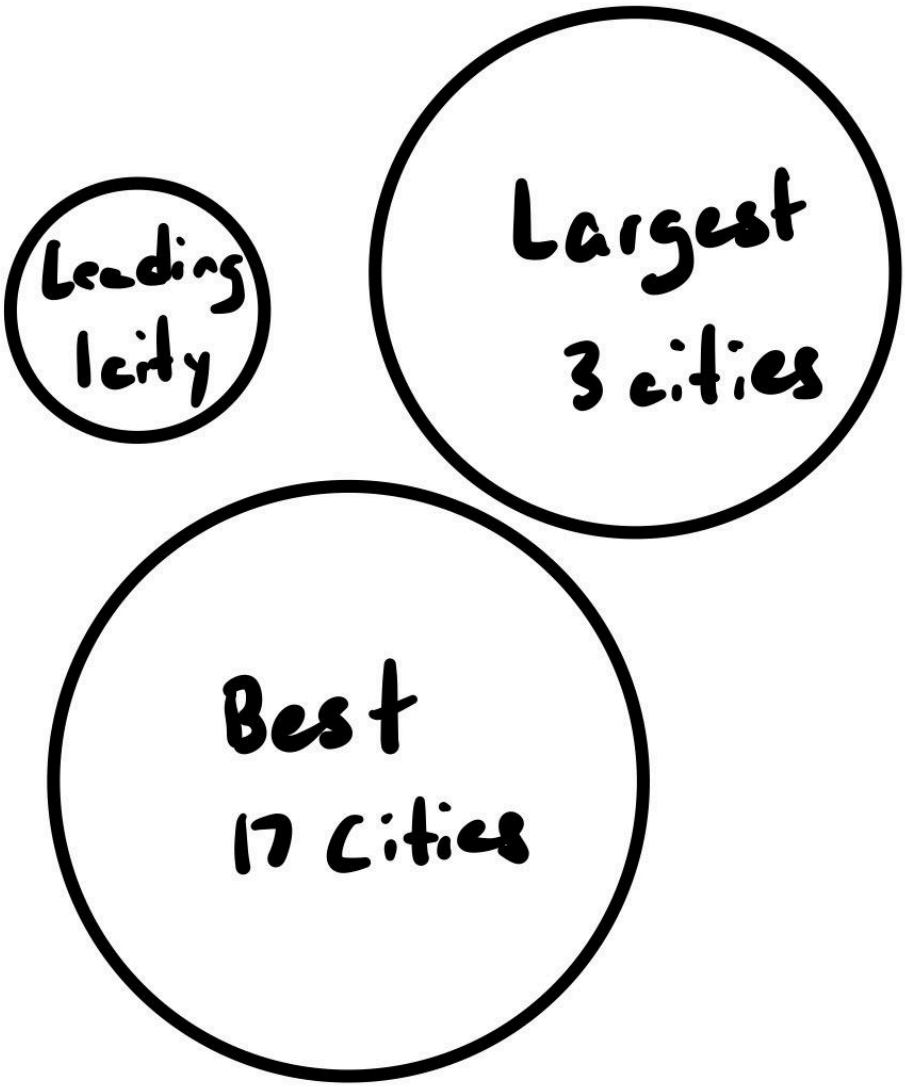
Description:
For a single city at a time, overlay the site's theme profile against the review theme profile across all the different categories that a city may advertise itself under. This gives us a quick visual read on whether the city pitch matches tourists' experiences.

Legend:
Advertised vs. Reviewed Scores (Color - Categorical) - We use color to categorize whether or not we are showing the score for a particular genre

for comparing advertised vs. reviewed scores.
Distance from center (Position – Quantitative): How strongly that theme shows up

(Reviewed as Sketch #24) Sketch #2

Questions:
Are cities competing to make the same superlative claims about themselves?



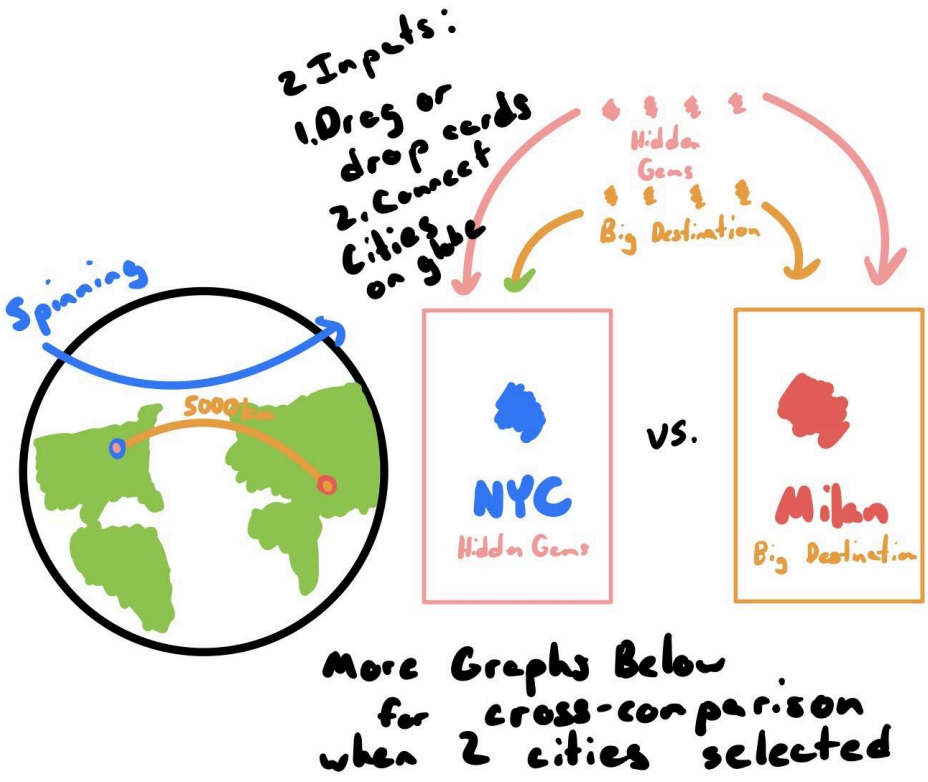
Description: The size of the bubble indicates the prominence and usage of a superlative across all the cities. This gives us an overview of the most utilized superlatives.

Legend:
Size of Bubble (Size - Circumference): The size of the bubble indicates the prominence of the superlative.

(Reviewed as Sketch #25) **Sketch #3**

Questions:
Are there cities that seem very different geographically but market themselves almost identically?

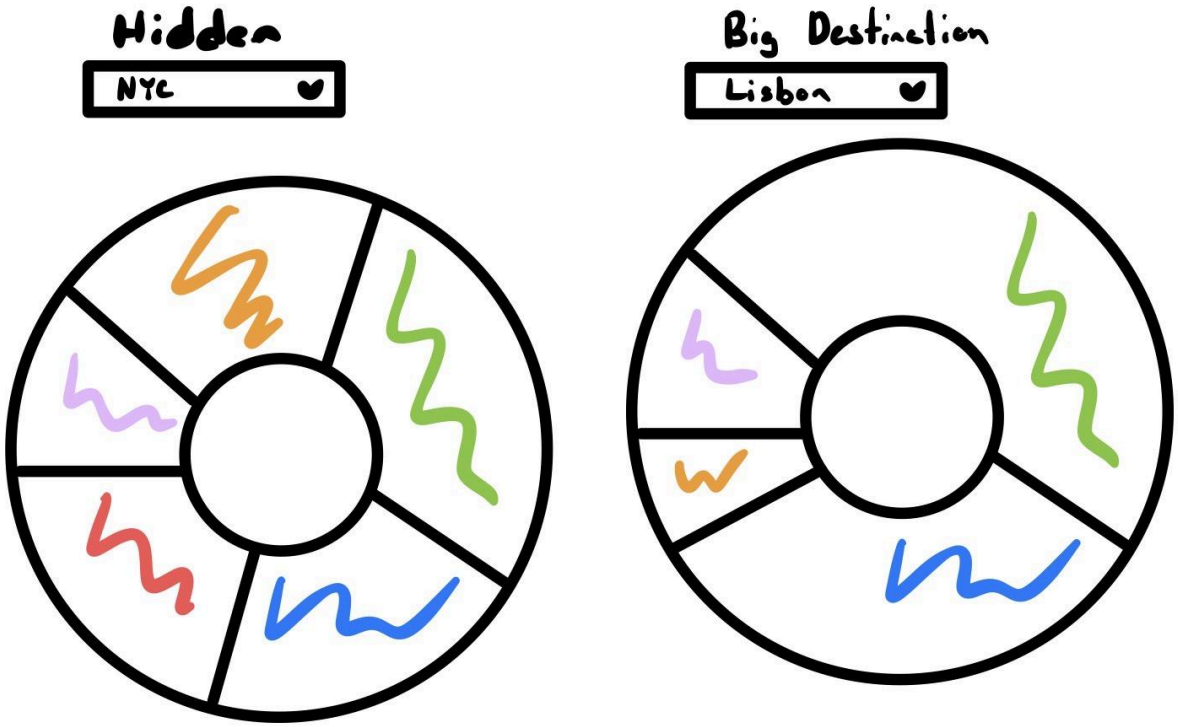
How do "hidden gem" cities market themselves differently from major tourist destinations? (Sort of)



Description:

To use this graphic, we can drag and drop cards or click two different cities to compare them on the globe.
The points on the globe and cards are also categorized by color so that we may compare gems to major cities.
When you select two cities, the system shows an in-depth cross-comparison between them.
Legend:
Border of Cards to Select From / City Central color on Globe (Color - Categorical): Categorizes cities between hidden gem and big destination
Point on Globe (Position - Categorical) - Geographical map point

(Reviewed as Sketch #26) Sketch #4
Question: How do "hidden gem" cities market themselves differently from major tourist destinations?

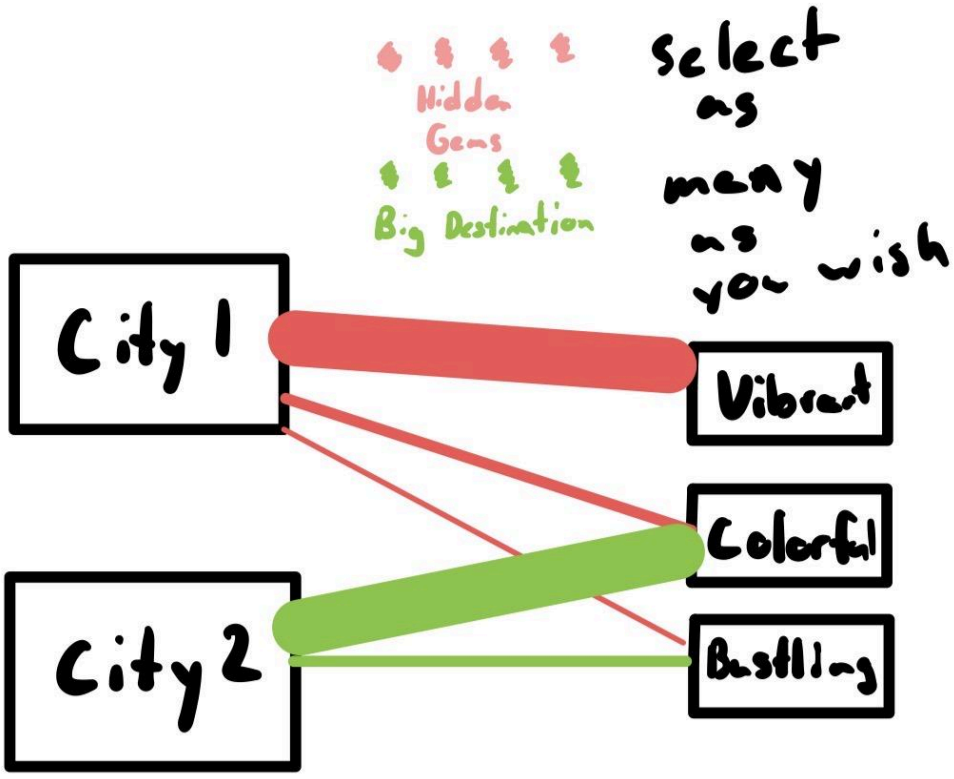


Description:
To compare each city's theme breakdown, allow us to have two dropdowns separated into hidden gem cities and big destination cities. Using the dropdowns, we can compare the thematic distribution breakdowns of 2

cities.

Legend:
Dropdown (Position - Categorical): We use positioned dropdowns to separate cities into hidden gems and big destinations.
Color on Circles (Color - Categorical): Using different colors on the wheels, we identify the most and least prominent themes within each selected city.
Wedge (Size - Quantitative): To show the magnitude of relevance for a theme based on city selection.

(Reviewed as Sketch #27) Sketch #5:
Question: Which cities use the most similar language overall? What words and phrases do cities use most often to market themselves?



Description:
Using the cards for each city above, we can see their distributional breakdowns of the adjectives they use to market themselves. We may

select as many cities as we wish to compare. This could also be done with reviews as well.

Legend:
Bar size coming from each card (Width - Size): Shows how prominent an adjective is to a city.
Border Color from each City's Card (Color - Categorical): Shows if a city is either a hidden gem or big destination.
Bar Color from each City's Card (Color - Categorical): Shows which bar belongs to which city.

Milestone 4

Question IDs

1. **What are major tourist cities trying to be known for? What words and phrases do cities use most often to market themselves?**
2. **Which cities use the most similar language to describe themselves?**
3. **Which cities have the most distinctive or unique marketing language?**
4. Do major tourist cities rely on the same adjectives and phrases to attract visitors?
5. **How might "hidden gem" cities market themselves differently from major tourist destinations?**
6. **What themes do cities emphasize most?**
7. Are cities competing to make the same superlative claims about themselves?
8. How do city marketing languages compare to actual reviews?
9. **Are there cities that seem very different geographically but market themselves almost identically?**
10. **Can we identify a city based only on the words and phrases used on its tourism website?**

Votes by Color:

Josh: Blue

Christine: Red

Nate: Yellow

(Reviewed as Sketch #_) ID	Question ID	Author	Vote
1, 6, 7, 13, 23, 26	1, 6, 5	XCZ, JZ	Yes Yes Yes
2	7	XCZ,	
3	3	XCZ,	
4, 9, 1	1, 9	JZ, NZ	
5, 8, 12, 16, 25	9	JZ, NZ, XCZ	Yes Yes Yes

11, 19	8	NZ	
14	2	XCZ	
15, 27	10	XCZ	Yes Yes Yes
17	1	XCZ	
21	1	NZ	
18	3	NZ	Yes Yes Yes
20	2	NZ	Yes Yes Yes
22	4	NZ	
23	7	JZ	
27	2	JZ	

Summary:

After reviewing our sketches, we found that the majority of the sketches converged into three primary categories: city breakdowns by personality traits, direct comparisons between cities, and overall trends found across cities.

A large share of our sketches converged on the same underlying idea: giving each city a distinct "personality" or thematic "fingerprint" based on the marketing themes it emphasizes most. These sketches share a common backbone, which is a per-city theme score that can be encoded as size, position, or color. After cleaning our dataset and assigning these theme scores to each city, the data validated this direction as cities cluster in visually distinct ways depending on which themes (food, nightlife, history, nature, etc.) dominate their marketing language, which means that simply encoding those scores visually should let a viewer spot cities with similar identities at a glance. This category of visualization is especially well-suited to our target audience of curious travelers, since it compresses a large amount of scraped textual data into an immediately readable form.

Second, we found that a handful of our graphs directly compare cities against each other. This is especially useful when users are deciding between two cities. Sketches 5, 8, 12, 16, and 25 all fall under this umbrella category.

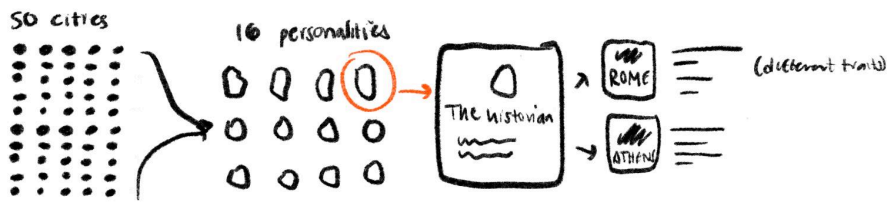
Lastly, some of the other cities fit under the very broad category of general trends by descriptor. Some examples of this are sketches 6 and 18. Some of these sketches fall under the other categories, so they will likely be used even if not necessarily for this exact idea. One final note is that we are open to incorporating other ideas if none of these work out, such as with sketch 20, which doesn't fit cleanly into any of the other categories.

Milestone 6

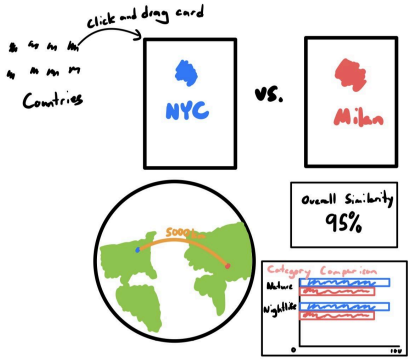
Visualizations

Our three main idea "buckets" are best summarized by Christine's sketch 1, the 16 personalities idea, Josh's sketch 5, the city one vs one idea, and Nate's sketch 10, which shows general trends between cities. The sketches are shown below again for convenience.

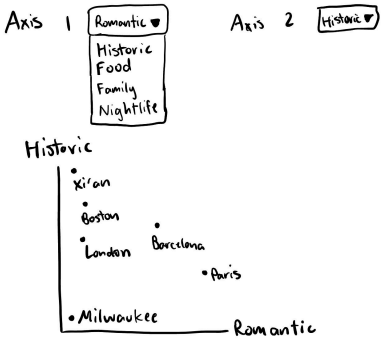
Sketch 1:



Sketch 5:



Sketch 10:



These three ideas will mostly serve to answer the following questions:

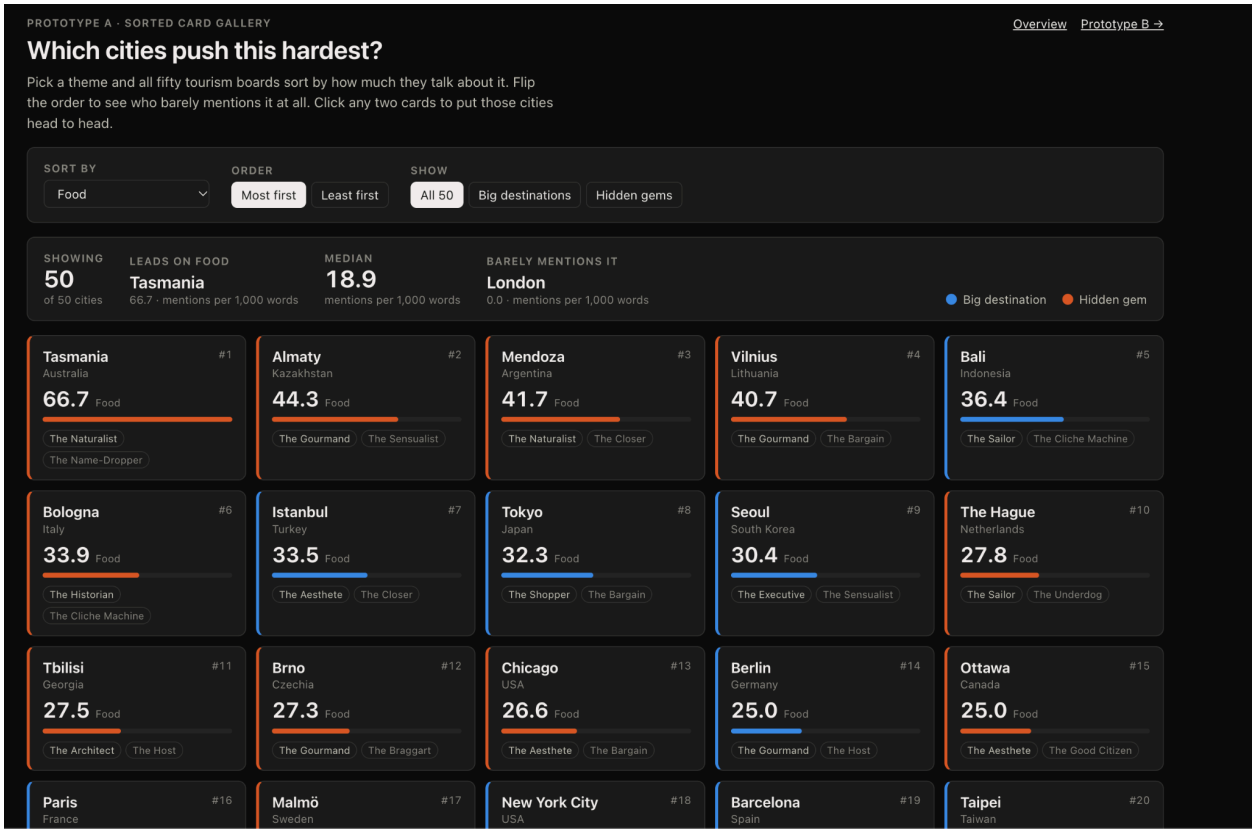
- Which cities use the most similar or unique language to describe themselves? **(answered by sketch 1)**
- Are there cities that seem very different geographically but market themselves almost identically? **(answered by sketch 5)**
- What themes or general trends do cities emphasize most? **(answered by sketch 10)**

These ideas were chosen from a combination of two main factors: (1) which questions would most help travelers who are trying to find new travel destinations, and (2) which ideas are most fun 🙂. We believe our idea could help travelers by allowing them to compare familiar cities to unfamiliar ones. They could, for example, choose the personality of a familiar city, and find others under the same category. By combining our personalities

and map ideas, travelers can easily find atmospherically-similar but geographically distant cities.

https://jz-17.github.io/interactive_destination_visualization/prototype-a/

This visualization is centered around a 3D, draggable globe. Each city is shown as a circle on the map. Clicking a city 1) adds the city to the selection panel on the right and 2) turns the globe so the city faces the viewer. Clicking a second city draws an arc between the pair and fills in every comparison view below, sort of like a geographic brush feature. The tables include the theme-by-theme diverging bars, the shared vocabulary, and the scatter plot with both cities ringed and directly labelled, as shown in our sketches. Selecting an archetype pills beside the globe dims the cities that do not match but leaves them drawn, on the globe and in the scatter, and still clickable. Hovering anything highlights it everywhere at once.



The Name-Dropper

Hamburg #36 Germany **13.5** Food The Aesthete · The Braggart	**Marseille** #37 France **13.0** Food The Sailor · The Understated	**Penang** #38 Malaysia **12.9** Food The Naturalist · The Name-Dropper	**Ohrid** #39 N. Macedonia **12.4** Food The Naturalist · The Understated	**Belgrade** #40 Serbia **11.9** Food The Aesthete · The Understated
Rome #41 Italy **10.8** Food The Historian · The Braggart	**Bucharest** #42 Romania **9.8** Food The Architect · The Understated	**Sao Paulo** #43 Brazil **8.1** Food The Gourmand · The Underdog	**Dubai** #44 UAE **7.8** Food The Historian · The Closer	**Prague** #45 Czech Republic **7.7** Food The Executive · The Cliche Machine
Cusco #46 Peru **7.0** Food The Historian · The Host	**Mecca** #47 Saudi Arabia **4.5** Food The Architect · The Braggart	**Marrakech** #48 Morocco **4.3** Food The Shopper · The Underdog	**Luxembourg City** #49 Luxembourg **3.7** Food The Historian · The Host	**London** #50 United Kingdom **0.0** Food The Historian · The Closer

The **#** on each card is its rank across all fifty, so it does not change when you filter to one category. Bar length is the city's value against the highest in the corpus.

Pick two cities to compare them head to head. Click one, then another — in any view on this page.

Holding **Luxembourg City**. Pick a second.

Theme measures are mentions per 1,000 words, so a long scrape and a short one are comparable. Source: official tourism-board homepage text, 50 cities.

HIDDEN GEM
Luxembourg City
Luxembourg · Europe

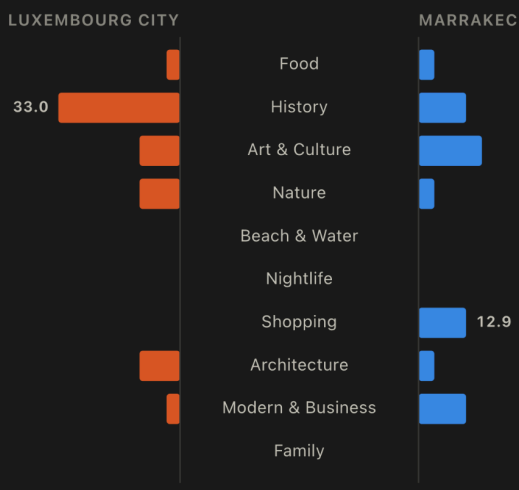
vs

Clear

BIG DESTINATION
Marrakech
Morocco · Africa

2,319 km apart · different regions

What they talk about



Mentions per 1,000 words. Labelled bars are each city's **archetype theme** — the one it over-indexes on, which is not always its longest bar.

How they sell it

Luxembourg City
The Host 1.46 sd Host · Underdog · Name-Dropper

Marrakech
The Underdog 1.35 sd Underdog · Good Citizen · Host

Words they share

No overlap in their top-8 distinctive words — these two pitch themselves with different vocabulary entirely.

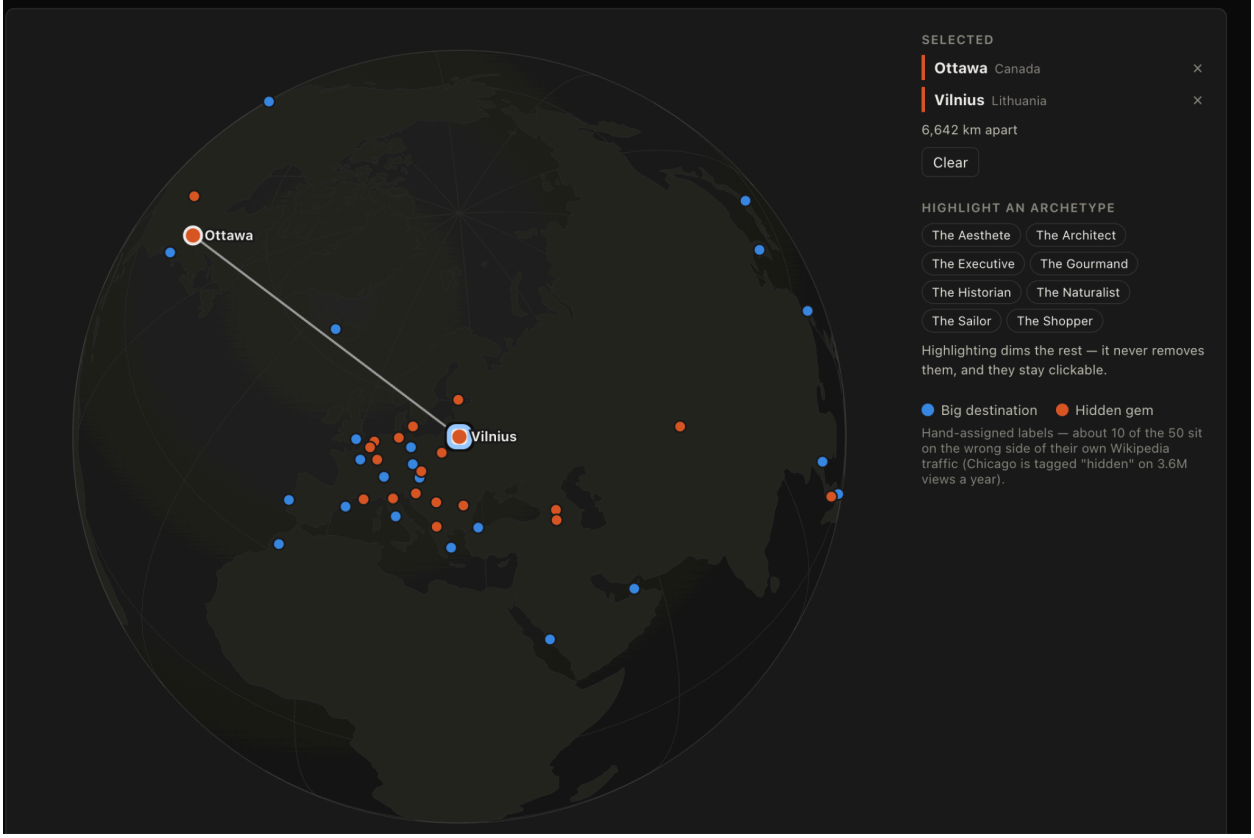
Signature word: **surprise** (Luxembourg City) · **oriental** (Marrakech)

https://jz-17.github.io/interactive_destination_visualization/prototype-b/

This visualization comprises a gallery of fifty city cards. A single row of controls sits above the cards. Picking a measure from the dropdown, like any of the ten themes, obscurity, rare words, cliché rate or personality strength, re-sorts all fifty cards right away. The summary row above them updates to show which city leads, the median, and which city barely mentions it. Most first and Least first flip the order. The three category pills (All 50/ Big destinations/Hidden gems) filter the cards. Cards that don't match are removed and the summary numbers are recalculated for whatever is left. Hovering a card highlights it. Clicking any two cards opens a head-to-head comparison below the gallery.

Pick two cities anywhere on Earth. Do they sound alike?

Spin the globe and click two cities. Everything below fills in for that pair: what each one talks about, how each one sells itself, and where both sit against the other forty-eight.





Comparison

The two prototypes differ in where you start. In the first prototype you spin a globe and click two cities, and every comparison fills in for that pair. It's the more convincing of the two because the globe makes our main finding physical, as over 50% of the cities have their closest-sounding twin in a different world region. In the second prototype, you pick a theme and all fifty cities re-sort into ranked cards. It's easy to read and shows the full range quickly of how themes vary across cities (ex: food in Tasmania has 66.7 mentions/1,000 words and London has 0). Each has a weakness. The second prototype is a long scroll and much less visually interesting, and because of the 3D layout of the first prototype, some cities get cut off.

Our next step is to build the final project around the first prototype and move the second prototype's sorted cards into a "browse everything" panel. We want a way to reach the hidden cities (maybe a count badge or a searchable list) and also create a sticky header in the card gallery so the sort measure stays in view. We also want to change the way we

measure how popular a city is because right now it's by Wikipedia traffic, which might not be very reliable.

Claude usage

We got together as a team to discuss our visualizations and which sketches we wanted to combine, alongside the interactions that link them. Based on this conversation, we wrote out some Claude prompts for the two prototypes, following the guidelines we were given in our labs. We then went through the iterative process of refining the sketches to ensure that our story was clear, and that no interactions were broken.

Milestone 7

Key insights:

- **Christine:**
    - A lot of cities claim to have the same characteristics. Tourism copy converges on a shared vocabulary. "Unique" itself is one of the most common words in it (26/50). Superlatives collapse onto best (27), largest (27), oldest (16).
    - The better known a city is, the fewer unique words it uses. For instance, London's vocabulary is 3.6% its own vs Marrakech 30.9%, and Ohrid 30.6%.
    - There are large gaps in themes. For Tasmania, nature is 111/1k, Bucharest history 56, The Hague water 56.
- **Nate:**
    - What cities leave out is as telling as what they say. For example, 28 of 50 sites never mention nightlife, 34 of 50 never mention families, and 20 of 50 never mention shopping. These are things many travelers actively look for, yet tourism boards mostly stay quiet about them. A "silence" view, showing gaps in each city's profile, could potentially be a good fit for our audience.
    - Food is the most-mentioned theme for 19 of 50 cities. Food, history, nature or art & culture is the top theme for 47 of 50. Only three cities lead with something else, and all three lead with beach & water.
- **Josh:**
    - Popular cities tend to brag about scale, citing crowd and visitor figures in the millions, as Dubai and Vienna do. Hidden cities brag about age instead: 8 of 25 cite a specific century, compared with none of the popular cities. Lacking fame, they use history as their proof of worth.
    - Ironically, the phrase "hidden gem" appears in only 4 of our 50 cities, and every one is a popular destination: London, Dubai, Berlin and Singapore. None of the 25 lesser-known cities use it. Famous places borrow the language of discovery to seem less obvious, while true hidden gems describe what they offer rather than how undiscovered they are.
    - Hidden cities pick a lane. Rather than trying to be everything to every traveler, they build their pitch around one clear identity, like Mendoza with its wine country or Tasmania with its wilderness. Popular cities can afford to cover food, shopping, history and nightlife all at once because their name already does the selling.

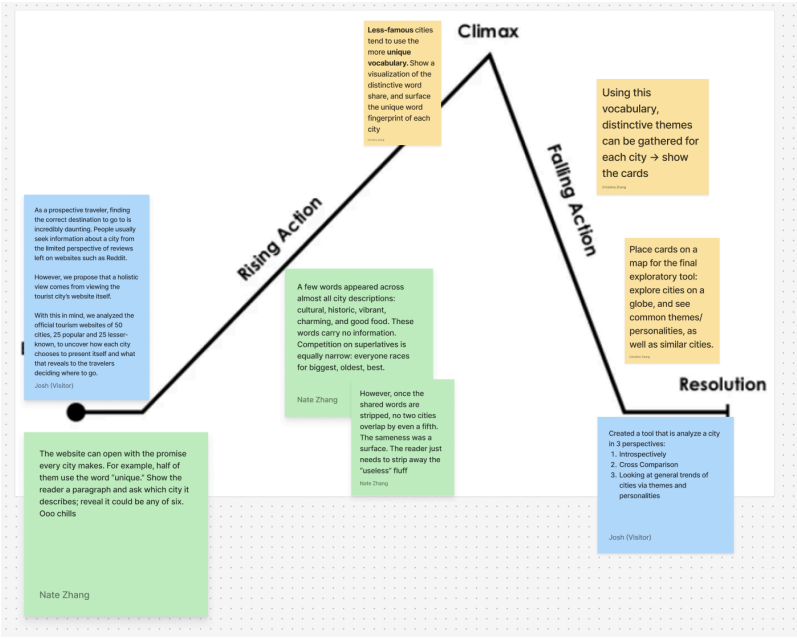
Main insight

The main message we decided to go with is that almost **every city claims to be one of a kind, but they nearly all say it with the set of words, and a city's language gets more generic the more visitors it gets.**

We chose this message because it pulls many of our other questions together, alongside the visualizations we've already sketched out. It covers many things we set out to explore: whether most cities use very similar language, whether the distinctive ones correlate to famous vs. hidden gem cities, and whether you can identify a city from its own words. It also explains something we didn't expect, which is that more than half of our cities sound most like a city on another continent. We thought this would be super interesting to visualize with our globe. In making this claim, we kept the wording careful on purpose so that we don't overstate the claim. We hope to run more analyses with different ways of categorizing famous vs. hidden gem cities (international visitors? Web searches? General online consensus?) to ensure the story is sound.

Data storyboard

We did our storyboard brainstorming on a FigJam board [link]



Prototype 1

Contributors: Christine Zhang, Nate Zhang, Joshua Zhang

2 Functional D3 Visualizations

https://jz-17.github.io/interactive_destination_visualization/prototype-a/

https://jz-17.github.io/interactive_destination_visualization/prototype-b/

Webpage

https://jz-17.github.io/interactive_destination_visualization/story/

Prototype A: Cards

A gallery of fifty city cards, with one row of controls above them. Picking a measure from the dropdown re-sorts all fifty cards. The summary row above the cards updates to show which city leads, the median, and which city barely mentions it. We also included filters for big destinations vs. hidden gems. Hovering a card highlights it. Clicking any two cards opens a head to head comparison below the gallery.

Prototype B: Globe

We used a globe instead of a flat map because the question underneath this prototype is about distance, and a rectangular projection distorts it. Clicking a city adds it to the selection panel and turns the globe so that city faces the viewer, so a click near the edge doesn't feel random. Clicking a second city draws an arc between the pair and fills in every comparison view below at once.

Site structure

The site is at /story/. It follows the storyboard we built in FigJam. The page will open on a paragraph of tourism copy with the city name removed and asks the reader to guess out of 6 options. Then it shows the shared words: 13 adjectives appear in 15 or more of our 50 cities, and 49 of the 50 use at least one. Twenty six cities call themselves "unique". In the rising action, we have 3 sections: Section 1 is what cities talk about (prototype A). Section 2 is how cities talk (personality), Section 3 is what cities leave out, which was a new sketch we came up with for this milestone. The main message will be Prototype B (the globe). 28 of 50 cities sound most like a city in another world region.
